exida

The manufacturer may use the mark:

CERTIFIED • IEC 61508 • SIL 3 CAPABLE • exida

Revision 3.1 April 25, 2024
Surveillance Audit Due
July 1, 2027

ANAB
ANSI National Accreditation Board
ACCREDITED
ISO/IEC 17065
PRODUCT CERTIFICATION BODY
#1004

Certificate / Certificat
Zertifikat / 合格証

ERD 1608037 C001

exida hereby confirms that the:

BM9 Series Slam Shut Valve

Emerson Process Management Regulator Technologies, Inc. China

Has been assessed per the relevant requirements of:

IEC 61508 : 2010 Parts 1-2

and meets requirements providing a level of integrity to:

Systematic Capability: SC 3 (SIL 3 Capable)

Random Capability: Type A Element

SIL 2 @ HFT=0; SIL 3 @ HFT = 1; Route 2_H

PFH/PFD_{avg} and Architecture Constraints must be verified for each application

Safety Function:
The Slam Shut Valve will move to the designed safe position within the specified safety time.

Application Restrictions:
The unit must be properly designed into a Safety Instrumented Function per the Safety Manual requirements.

exida Certification

Jack Gao

Evaluating Assessor

Raymond Lee

Certifying Assessor

Certificate / Certificat / Zertifikat / 合格証

ERD 1608037 C001

Systematic Capability: SC 3 (SIL 3 Capable)

Random Capability: Type A Element

SIL 2 @ HFT=0; SIL 3 @ HFT = 1; Route 2_H

PFH/PFD$_{avg}$ and Architecture Constraints must be verified for each application

BM9 Series

Slam Shut Valve

Systematic Capability:

The product has met manufacturer design process requirements of Safety Integrity Level (SIL) 3. These are intended to achieve sufficient integrity against systematic errors of design by the manufacturer.

A Safety Instrumented Function (SIF) designed with this product must not be used at a SIL level higher than stated.

Random Capability:

The SIL limit imposed by the Architectural Constraints must be met for each element. This element meets *exida* criteria for Route 2_H.

IEC 61508 Failure Rates in FIT (FIT = 1 failure / 10^9 hours)

BODY SIZE (DN)	END CONNECTION STYLE	Slam Shut Controllers
200 250 300	NPS 8 NPS 10 NPS 12 ASME CL300 ASME CL600	OS9/80X-BP-R OS9/80X-BPA-D-R OS9/80X-MPA-D-R OS9/80X-APA-D-R OS9/84X-R OS9/88X-R OS9/80X-PN-R OS9/84X-PN-R
Options:		
Proximity switch		
Electrovalve - Must meet the requirements of IEC 61511, Part 1, Para 11.5 to be used in a safety system		
IT/3V three-way valve for setting control (Pe max 50 bar)		

Application/Device/Configuration	λ_S	λ_{DU}
Valve, Full Stroke, Clean Service	66	542
Piston Controllers - OPSO	126	89
Piston Controllers - UPSO	102	99
Piston Controllers - OUPSO	126	99
Diaphragm Controllers - OPSO	181	94
Diaphragm Controllers - UPSO	156	115
Diaphragm Controllers - OUPSO	181	115

SIL Verification:

The Safety Integrity Level (SIL) of an entire Safety Instrumented Function (SIF) must be verified via a calculation of PFH/PFD$_{avg}$ considering redundant architectures, proof test interval, proof test effectiveness, any automatic diagnostics, average repair time and the specific failure rates of all products included in the SIF. Each element must be checked to assure compliance with minimum hardware fault tolerance (HFT) requirements.

The following documents are a mandatory part of certification:

Assessment Report: ERD 23/12-091 R001 V1 R1 (or later)

Safety Manual: D104516X012 BM9 Safety Manual, Apr 2021 (or later)

exida®
80 N Main St
Sellersville, PA 18960